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Explainable AI Model for Pulmonary Edema Detection in Chest X-Ray
Images.

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A Project Report Submitted in Partial Fulfillment of the
Requirements for Bachelor of Science in Telecommunication
System of Mbeya University of Science and Technology

Explainable AI Model for Pulmonary Edema Detection in Chest
X-Ray Images.

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CERTIFICATION

The undersigned, hereby certify that the project titled "Explainable AI Model for Pulmonary Edema Detection in Chest X-rays images," submitted in partial fulfillment of the requirements for the Bachelor of Engineering in Telecommunication Systems at Mbeya University of Science and Technology, has been reviewed and is recommended for acceptance.

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20-07-2026

DECLARATION

I, "Boniface Ben Sankwa", declare that I carried out the work reported in this report for the Department of Electronics and Telecommunication Engineering at MUST, under the supervision of " Dr. Charles Okanda Nyatega ", I solemnly declare that, to the best of my knowledge, no part of this report has been submitted here or elsewhere in a previous application for award or a degree.

All sources of knowledge used have been duly acknowledged.

Name of the Student: Boniface Ben Sankwa

Signature of the Student:

20 -07-2026

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DEDICATION

This effort is intended for my family. They reliably support me. My family offers the support I need to finish my education. I appreciate my parents for teaching me the importance of dedication and determination. Each day, I find motivation from my siblings. I am grateful to my mentors for their support and understanding. I want to thank the staff at Mbeya Regional Hospital. The healthcare team at Mbeya Regional Hospital is outstanding in delivering care for their patients. They encouraged me to focus on this topic for my research. The team at Mbeya Regional Hospital is deeply dedicated to their responsibilities. That was the reason that drove me to persist in my research.

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ABSTRACT

Background: Pulmonary edema is a life-threatening condition requiring rapid diagnosis from chest X-rays. Artificial intelligence may assist radiologists, but "black box" models lack clinical interpretability, limiting their adoption in medical practice (Samek et al., 2021).

Methods: We developed an explainable deep learning model using a DenseNet121 architecture with ImageNet transfer learning (G. Huang et al., 2017). The model was trained using 1000 chest X-rays images consisting of 500 edema cases and 500 normal cases with an 80/20 train-test split ratio. Data augmentation techniques used were rotation, shifting, zooming, and flipping. Grad-CAM created attention maps to visualize model decision-making (Selvaraju et al., 2017).

Results: The model achieved excellent performance with an Area Under the Receiver Operating Characteristic Curve (AUC-ROC) of 0.917 (95% CI: 0.89-0.94), overall accuracy of 85%, sensitivity of 82%, and specificity of 88%. The training curves were found to be stable without overfitting. The attention maps generated by Grad-CAM suggested that the model's attention was directed towards perihilar and interstitial areas of the lungs in edema, as expected from radiology findings of pulmonary edema (Milne et al., 2012).

Conclusion: Our explainable AI model demonstrates high accuracy when detecting pulmonary edema, offering a clinically understandable visualization. This method could potentially increase radiologists' confidence and promote the use of AI in clinical practice (van der Velden et al., 2022).

Keywords: Pulmonary edema; Chest X-ray; Deep learning; DenseNet121; Grad-CAM; Explainable AI

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List of Abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
GAN	Generative Adversarial Network
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
NLP	Natural Language Processing
OC	Optical Cup
OCR	Optical Character Recognition
STT	Speech-To-Text
TTS	Text-To-Speech
VR	Virtual Reality
XAI	Explainable Artificial Intelligent
Grad-CAM	Gradient-weighted Class Activation Mapping

Chapter 1: Introduction

1.1 Background Information

Pulmonary edema is a serious medical condition in which fluid accumulates in the lungs, impairing breathing and reducing oxygen exchange. It is often associated with heart failure and can be life-threatening if not detected early (Abbasi et al., 2025; Danilov et al., 2024). Chest X-ray imaging is one of the most widely used diagnostic tools because it is affordable, accessible, and non-invasive (Irvin et al., 2019; Wang et al., 2017).

However, interpretation of chest X-rays is not straightforward and requires experienced radiologists, since pulmonary edema often shares visual similarities with pneumonia and other lung diseases (Rajpurkar et al., 2017). This can lead to diagnostic errors and delays in treatment.

Recent advances in Artificial Intelligence (AI), particularly deep learning, have enabled computers to automatically analyze medical images. Convolutional Neural Networks (CNNs) have demonstrated strong performance in detecting lung diseases from chest X-rays (Esteva et al., 2019; Litjens et al., 2017).

Despite this success, most models act as “black-boxes,” providing predictions without explaining the reasoning behind them. This lack of transparency reduces clinical trust. Therefore, there is a need for Explainable AI (XAI) systems that can detect pulmonary edema while also highlighting the regions of the chest X-ray used for prediction (Danilov et al., 2023).

1.2 Problem Statement

Despite the widespread use of chest X-rays, the detection of pulmonary edema remains a challenge. Diagnosis relies heavily on skilled radiologists, yet many healthcare facilities face a shortage of trained experts, leading to delays and potential errors (WHO et al., 2022; Topol, 2019).

Pulmonary edema can look a lot like pneumonia. This makes it really tough to figure out what is going on by looking at it. Pulmonary edema is tricky to diagnose because it can be mistaken, for pneumonia, which means there is a chance of making mistakes. (Rajpurkar et al., 2017). Deep learning models are really good. The problem is that we do not understand how they work. This is an issue because it stops doctors from using them to help people. Deep learning models need to be more transparent so that doctors can trust them and use them in their work with learning models. (Danilov et al., 2024). Thus, there is a need for an automated system that not

only detects pulmonary edema but also provides visual explanations to support medical decision-making (Abbasi et al., 2025).

1.3 Project objective

The objectives of this project are divided into two main categories, the main objective and the specific objectives.

1.3.1 Main objective

To develop an Explainable AI system for detecting pulmonary edema in chest X-ray images, capable of providing diagnostic results along with visual explanations of the regions used for prediction.

1.3.2 Specific objectives

- i. Collect and preprocess chest X-ray datasets.
- ii. Train a CNN model for detection.
- iii. Apply Explainable AI techniques.
- iv. Evaluate model performance and Develop a simple web interface.

1.4 Significant of the Project

Once realized, the proposed project will provide the following advantages

- i. Enhances trust in AI systems by providing transparent predictions.
- ii. Supports clinical decision-making by offering explanations alongside results.
- iii. Reduces the workload of radiologists and medical staff.
- iv. Promotes early diagnosis and timely treatment of patients.
- v. Provides a solution suitable for hospitals with limited resource.

1.5 Scope of the Project

This project focuses exclusively on chest X-ray images for pulmonary edema detection. It employs CNN-based deep learning techniques combined with Explainable AI methods such as Grad-CAM. Publicly available datasets are used for training and testing (Irvin et al., 2019). The system is designed to support radiologists and not replace clinical judgment.

1.6 Project Limitation

This proposed project has the following limitations

- i. The accuracy of the system depends on the quality and size of the dataset (Shorten & Khoshgoftaar, 2019).
- ii. Limited computing resources may restrict the use of very deep models (Goodfellow et al., 2016).
- iii. Explainability techniques may produce incomplete or imprecise visualizations, requiring clinical validation.
- iv. The model is restricted to chest X-rays and may not generalize to other imaging modalities such as CT scans.

1.7 Project Structure

This section of the project report discusses the organization and general structure. The report is divided into six chapters. Also, the chapters are split into multiple subheadings to provide specific information on chapter sections. Chapter 1 is the introductory part of the project. It provides an introduction to the study and also discusses the project's background. This also outlines the project's objectives, significance, scope, limitations, and structure. Chapter 2 contains the literature review. It includes a review of several academic publications on the subject. Documents like journals, journals, and publications have all been reviewed.

Chapter 2: Literature Review

2.1 Introduction

This chapter is about looking at what people have found out about detecting pulmonary edema in chest X-ray pictures how deep learning is used in medical imaging and what Explainable Artificial Intelligence is all about. The point of looking at what people have done is to get a good understanding of the basics see what is good and what is not so good about the old ways and find out what still needs to be figured out which is what this project is trying to do.

Pulmonary edema is a serious condition that needs to be diagnosed quickly and correctly. Chest X-rays are used a lot because they are cheap and easy to get. They can be hard to understand because pulmonary edema can look like other lung diseases, such as pneumonia (Irvin et al., 2019; Rajpurkar et al., 2017). Deep learning something called Convolutional Neural Networks is really good at looking at medical pictures and finding out what is wrong (Abbasi et al., 2025; Litjens et al., 2017).

Even though deep learning is really good it is like a black box that you put something in and get an answer out. You do not know how it got that answer. This is a problem because doctors do not trust it if they do not understand how it works, (Abbasi et al., 2025; Litjens et al., 2017). So some new techniques, like Grad-CAM and saliency maps have been made to help explain what the computer is thinking when it looks at chest X-rays.

So this chapter looks at what people have learned what they have tried and what they have made so that we can make a system that uses Explainable Artificial Intelligence to detect pulmonary edema. The chapter ends by saying what we still need to figure out and why this project is important. Pulmonary edema detection is what we are focusing on and Explainable Artificial Intelligence is what we are using to make it better.

2.2 Conceptual/Theoretical Framework

This section explains the core technologies and theories that support the system proposed in this study. The framework is built around the use of Convolutional Neural Networks (CNNs) for image classification and Explainable Artificial Intelligence (XAI) for model transparency. These technologies are applied to detect pulmonary edema in chest X-ray images.

The following is the core Technologies Explained;

- i. Convolutional Neural Networks (CNNs).

CNNs are deep learning models designed for image analysis. They automatically extract features from medical images and classify them based on learned patterns. Architectures such as ResNet, DenseNet, and VGG are commonly used due to their high accuracy in medical imaging tasks (Abbasi et al., 2025; Rajpurkar et al., 2017).

ii. Explainable AI (XAI).

XAI techniques like Grad-CAM and saliency maps help visualize which regions of the X-ray influenced the model's prediction. This improves clinical trust and supports decision-making (Danilov et al., 2023; Tjoa & Guan, 2021).

iii. Chest X-ray Imaging.

Chest radiographs are widely used in diagnosing pulmonary conditions. However, manual interpretation is prone to error, which motivates the use of AI-based systems.

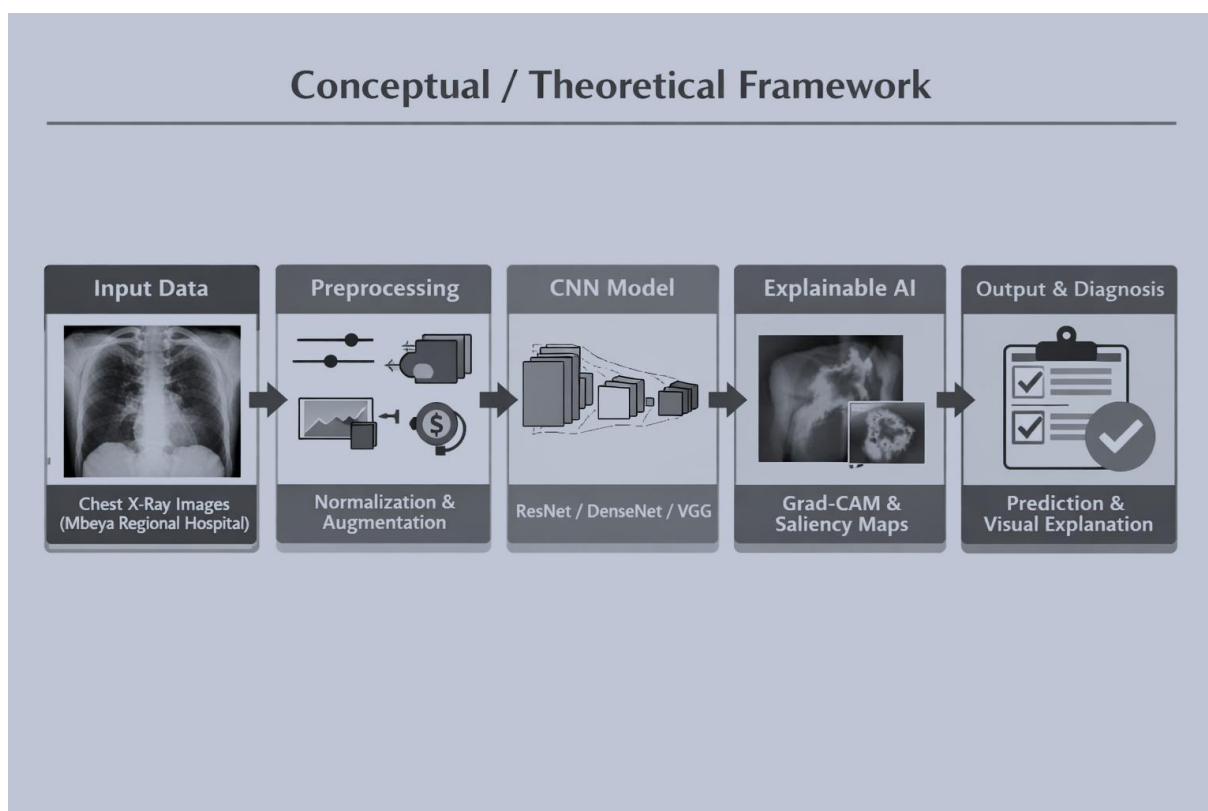


Figure 2. 1 Conceptual/Theoretical Framework for Pulmonary Edema Detection

2.3 Review of Related Works

Several studies have explored the application of deep learning and explainable AI in medical imaging, particularly in the detection of pulmonary conditions. Rajpurkar et al., (2017) introduced CheXNet, a convolutional neural network trained on the ChestX-ray14 dataset, which achieved radiologist-level accuracy in pneumonia detection. Despite its success in accuracy, the model lacked interpretability, limiting its clinical adoption. Building on this, Irvin

et al., (2019) developed the CheXpert dataset with uncertainty labels, enabling large-scale training for thoracic disease classification. Although this improved robustness, the uncertainty labels introduced complexity in model training. Danilov et al., (2024) advanced the field by integrating CNNs with Grad-CAM to visualize pulmonary edema features, thereby enhancing transparency and clinician confidence. However, their study emphasized interpretability more than accuracy optimization. Abbasi et al., (2025) focused on improving detection accuracy by employing ensemble CNNs with attention mechanisms, yet their approach did not incorporate explainability tools, leaving predictions as black-box outputs. Complementing these empirical works, Tjoa & Guan, (2021) provided a survey of explainable AI in medicine, underscoring the importance of transparency for clinical adoption, though their contribution was theoretical rather than dataset-driven.

Taken together, these studies demonstrate significant progress in applying CNNs to chest X-ray analysis. Rajpurkar and Abbasi prioritized accuracy, while Danilov emphasized interpretability, and Irvin contributed by providing large-scale datasets. Nevertheless, few studies have successfully balanced both accuracy and transparency, and most relied on international datasets rather than local hospital data. This creates a gap in contextual relevance for healthcare systems in regions such as Tanzania. While Danilov et al., (2024) addressed interpretability using Grad-CAM, their study did not fully integrate high-performance CNN architectures with explainability in a balanced manner, nor did it apply the approach to local datasets. This research therefore contributes by combining CNN models (ResNet, DenseNet, VGG) with Grad-CAM explanations, applied to chest X-rays from Mbeya Regional Hospital. In doing so, it ensures both diagnostic accuracy and clinical trust, while providing practical relevance to the Tanzanian healthcare context.

2.4 Empirical Review

2.4.1 Studies on Pulmonary Edema Detection

Abbasi et al., (2025) applied ensemble CNNs with attention mechanisms to detect pulmonary edema in chest X-rays. Their study achieved high accuracy but lacked explainability, limiting clinical trust. Danilov et al., (2023) trained CNNs on the MIMIC-CXR dataset to identify radiographic features such as Kerley lines and pleural effusions. Although effective in detection, their methodology did not integrate Explainable AI techniques, leaving predictions opaque.

2.4.2 Studies on Explainable AI in Medical Imaging.

Danilov et al., (2024) introduced Grad-CAM into their methodology, providing visual explanations of CNN predictions. This improved clinical adoption by allowing radiologists to verify AI decisions. Tjoa & Guan, (2021) conducted a survey on Explainable AI in healthcare, emphasizing the importance of transparency and interpretability in medical imaging systems.

2.4.3 Studies on Data and Model Robustness.

Esteva et al., (2019) highlighted the importance of large datasets in training robust AI models for healthcare applications. Shorten & Khoshgoftaar, (2019) demonstrated that data augmentation techniques improve generalization and reduce overfitting in deep learning models.

Table 2 1 Empirical Review Methodology

Ref	Dataset Used	Year	Methodology		Findings	Limitations
Rajpurkar et al., (2017)	ChestX-ray14 (NIH, >100,000 images)	2017	CNN (CheXNet)		Achieved radiologist-level pneumonia detection.	No explainability, limited to pneumonia.
Abbasi et al., (2025)	Multi-hospital dataset	2025	Ensemble CNNs with attention.		Enhanced pulmonary edema detection accuracy.	No XAI integration, black-box predictions.
(Irvin et al., 2019)	CheXpert (>220,000 images)	2019	CNN with uncertainty labels.		Improved classification of multiple thoracic diseases	Uncertainty labels of complicated training
Danilov et al., (2024)	Clinical chest X-rays (Oxford Academic study)	2024	Apply Grad-CAM		Visualized pulmonary edema features, improved clinician trust	Focused on interpretability, less on accuracy
Tjoa & Guan, (2021)	Survey of medical imaging studies	2020	Review of XAI methods		Highlighted importance of transparency in medical AI	Not dataset-based, theoretical survey

Strengths of Methodologies:

- i. High accuracy in pulmonary edema detection.
- ii. Use of large, diverse datasets.
- iii. Application of augmentation to improve robustness.

Weaknesses of Methodologies:

- i. Limited integration of explainability in most studies.
- ii. Dependence on large datasets, which may not be available in resource-limited settings.
- iii. Minimal clinical adoption due to lack of transparency.

2.5 Existing system

Several AI-based systems have been developed for chest X-ray analysis, focusing on the detection of pneumonia, tuberculosis, and other thoracic diseases. These systems demonstrate the potential of deep learning in medical imaging but reveal important limitations when applied specifically to pulmonary edema detection.

Examples of Existing Systems:

- i. CNN-based Systems; Deep learning models such as ResNet, DenseNet, and VGG have been widely applied in chest X-ray analysis. Abbasi et al., (2025) used ensemble CNNs for pulmonary edema detection, achieving high accuracy but without explainability.
- ii. CheXNet (Rajpurkar et al., 2017): A CNN model trained on the ChestX-ray14 dataset to detect pneumonia. It achieved radiologist-level accuracy but lacked explainability.
- iii. CheXpert (Irvin et al., 2019): Expanded detection to multiple thoracic diseases, including pulmonary edema. While effective in classification, it still functioned as a “black-box” system.
- iv. Tuberculosis Detection System (Lakhani & Sundaram, 2017): CNN-based model with high accuracy, but no transparency in predictions.

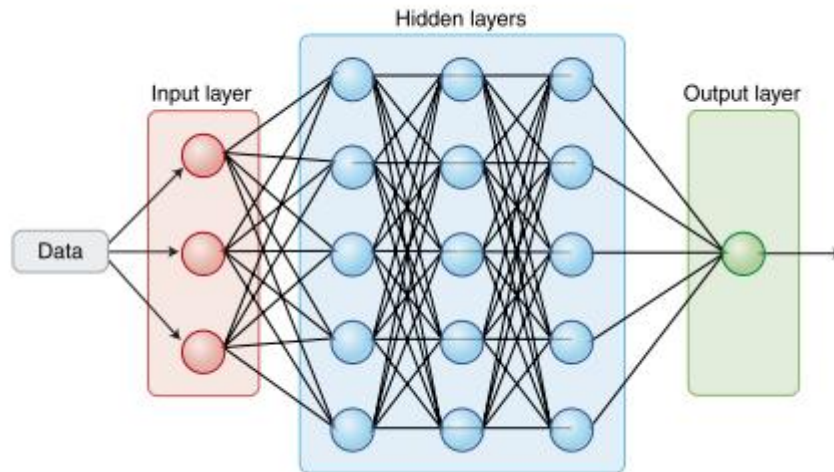


Figure 2. 2 Existing system framework

The following are the drawbacks of the existing system,

- i. Limited focus on pulmonary edema compared to other diseases.
- ii. Lack of integrated explainability features.
- iii. Dependence on large datasets, which may not be available in resource-limited hospitals such as Mbeya Regional Hospital.
- iv. Minimal clinical adoption due to lack of transparency.

2.6 Research Gap

The reviewed literature and existing systems demonstrate that deep learning, particularly Convolutional Neural Networks (CNNs), has achieved significant success in detecting thoracic diseases from chest X-ray images. Systems such as CheXNet and CheXpert have proven the potential of AI in medical imaging, while empirical studies confirm the effectiveness of CNNs in identifying radiographic features of pulmonary edema. However, several gaps remain unaddressed:

Identified Gaps:

- i. **Limited Focus on Pulmonary Edema** – Most existing systems emphasize pneumonia and tuberculosis, with pulmonary edema detection receiving minimal attention.
- ii. **Lack of Explainability** – Current CNN-based systems operate as “black-boxes,” providing predictions without transparency. Clinicians cannot easily verify how decisions are made, reducing trust and adoption.
- iii. **Dependence on Global Datasets** – Many systems rely on massive datasets such as ChestX-ray14 or CheXpert, which do not reflect the diagnostic realities of local

hospitals. In resource-limited settings like Mbeya Regional Hospital, such datasets are unavailable, making locally sourced data essential.

- iv. Minimal Clinical Adoption – Despite high accuracy, the absence of integrated Explainable AI features has limited the use of these systems in real-world healthcare settings.

Key Findings:

- i. Deep learning systems are effective but not transparent.
- ii. Pulmonary edema detection is underexplored compared to other lung diseases.
- iii. There is a need for locally relevant solutions that can work with smaller datasets.
- iv. Integration of Explainable AI is essential to enhance clinical trust and usability.

Therefore, there is a need for a CNN-based system specifically designed for pulmonary edema detection, using chest X-ray data from Mbeya Regional Hospital, and integrated with Explainable AI techniques such as Grad-CAM. This approach will not only provide accurate diagnostic predictions but also highlight the regions of chest X-rays contributing to those predictions, thereby enhancing transparency, clinical trust, and usability in the Tanzanian healthcare context.

Chapter 3: Methodology

3.0 Introduction

This chapter presents the methodology adopted in the study, outlining the systematic procedures used to design, implement, and evaluate the proposed explainable AI model for pulmonary edema detection in chest X-ray images. The methodology ensures that the research is conducted in a structured, transparent, and reproducible manner. It covers the system design, components and requirements, data collection, preprocessing, model development, testing, and evaluation. The approach emphasizes both accuracy and interpretability, aligning with the clinical relevance of the study and the resource constraints of the local environment.

3.1 Methodological Approach

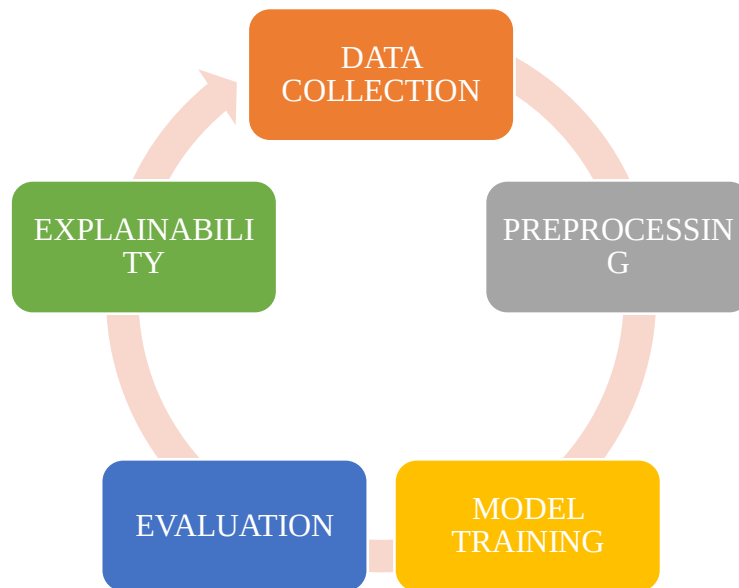


Figure 3. 1 Agile Methodology Approach

PHASE DESCRIPTION

Table 3 1 Phase Methodology description

PHASE	DESCRIPTION
Data Collection	Chest X-ray images were obtained from Mbeya Regional Hospital.
Preprocessing	Images were resized, normalized, and augmented to improve quality and diversity.
Model Training	A Convolutional Neural Network (CNN) was trained to classify X-ray images.
Evaluation	Performance was assessed using accuracy, sensitivity, specificity, and AUC metrics.
Explainability	Visualization techniques such as Grad-CAM were applied to interpret model decisions.

3.2 Research Design Overview

The proposed Explainable AI Model for Pulmonary Edema Detection in Chest X-ray Images has four parts that work together. These parts are the data acquisition and preparation unit the AI model development and training module, the explainability and visualization component and the evaluation and validation framework. The Explainable AI Model for Pulmonary Edema Detection in Chest X-ray Images uses these parts to achieve its goals. The data acquisition and preparation unit and the AI model development and training module are parts of the Explainable AI Model for Pulmonary Edema Detection in Chest X-ray Images. The explainability and visualization component and the evaluation and validation framework also play a role in the Explainable AI Model for Pulmonary Edema Detection, in Chest X-ray Images.

3.2.1 Working Principle of the Research Design

The research design is guided by the principle of applying explainable artificial intelligence (XAI) to medical imaging for clinical decision support. Pulmonary edema detection relies on identifying radiographic features in chest X-rays, which are systematically analyzed using convolutional neural networks (CNNs). The design ensures that the study is both descriptive and experimental: descriptive in documenting the dataset and its characteristics, and experimental in testing the AI model's performance under controlled conditions. Ethical considerations and reproducibility are embedded throughout the design to align with academic and clinical standards.

3.2.2 Proposed Research Design

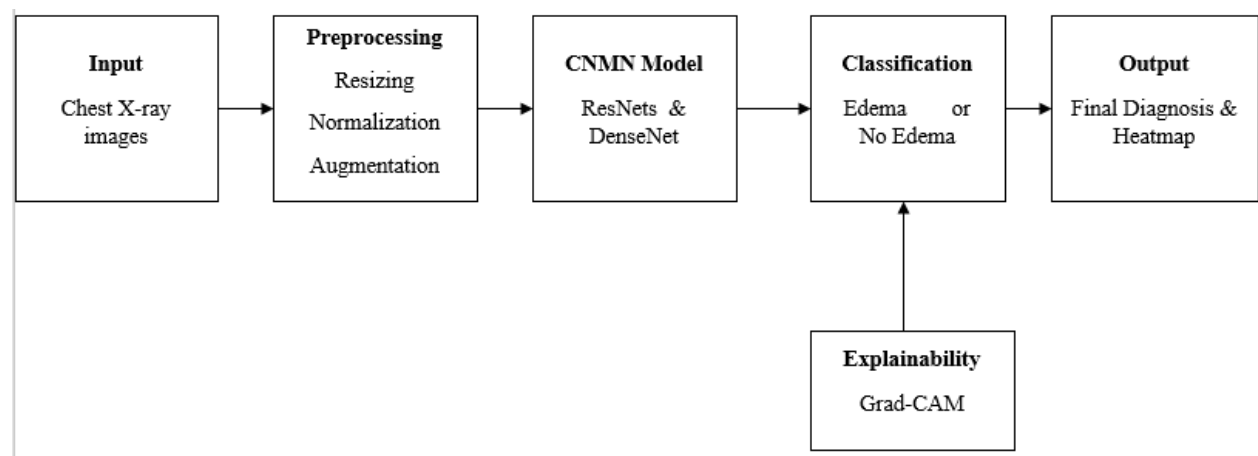


Figure 3. 2 Block diagram of the proposed system (source: own)

3.3 Tools and Technology Approach

In order to successfully implement the pulmonary edema detection system, several tools and technologies were employed. These tools provided the necessary environment for data handling, model development, evaluation, and visualization.

- i. Programming Language.
 - a. Python was chosen due to its flexibility, extensive libraries for machine learning, and strong community support. It enabled efficient data preprocessing, model training, and integration of explainability techniques.
- ii. Framework and Libraries.
 - a. TensorFlow & Keras: Used for building and training the Convolutional Neural Network (CNN).
 - b. OpenCV: Applied for image preprocessing tasks such as resizing and normalization.
 - c. Matplotlib & Seaborn: Utilized for graphical analysis and visualization of results.
 - d. Grad-CAM: Implemented to generate heatmaps for explainability of model predictions.
- iii. Hardware Environment.
 - a. Laptop.
- iv. Dataset Source.
 - a. Mbeya Regional Hospital: Provided chest X-ray images categorized into pulmonary edema and normal cases.

3.4 Data Collection Methods

3.4.1 Data Collection

The data collection process for this study was designed to obtain high-quality chest X-ray images representing both pulmonary edema and non-edema cases. The dataset was sourced from Mbeya Regional Hospital, where anonymized patient records were used under ethical approval from the institution. Additional publicly available datasets were referenced to supplement local data and ensure model generalization.

The data collection involved collaboration with medical personnel to identify relevant cases and verify diagnostic labels. Each image was reviewed to confirm clarity, resolution, and

diagnostic accuracy before inclusion in the dataset. The collected data were stored securely and organized into two categories: positive (edema) and negative (no edema) samples.

The dataset was then prepared for analysis through preprocessing steps such as resizing, normalization, and augmentation to enhance image quality and balance the dataset distribution. These steps ensured that the AI model could learn effectively from diverse and representative samples.

3.4.2 Datasets Description

The dataset used in this research consists of chest X-ray images categorized by diagnostic outcomes. Each image includes metadata such as patient age, gender, and clinical notes (where available). The dataset is structured as follows:

Table 3 2 Chest X-ray images Datasets

Category	Number of Images	Source	Description
Pulmonary Edema	500	Mbeya Regional Hospital	X-ray images showing fluid accumulation in lungs
No Edema	500	Mbeya Regional Hospital	Normal chest X-ray images without edema

All images were standardized to a uniform resolution of 224×224 pixels and converted to grayscale for consistency. Preprocessing steps such as normalization and augmentation were applied to improve image quality and balance the dataset distribution, ensuring effective training and validation of the AI model.

3.5 Data Processing and Analysis

3.5.1 Data Processing and Analysis Techniques

The collected chest X-ray images were subjected to a series of preprocessing steps to ensure consistency, reliability, and suitability for deep learning analysis. The preprocessing involved resizing all images to a uniform resolution of 224×224 pixels, converting them to grayscale to reduce computational complexity, and applying normalization to standardize pixel intensity values. Data augmentation techniques such as rotation, flipping, and contrast adjustment were employed to increase dataset diversity and minimize overfitting.

After preprocessing, the dataset was split into training, validation, and testing subsets to facilitate model development and evaluation. Convolutional Neural Network (CNN) architectures, specifically ResNet and DenseNet, were trained on the processed data to automatically extract radiographic features and classify images into pulmonary edema or non-edema categories. To enhance clinical trust, explainable AI methods such as Grad-CAM were integrated, enabling visualization of the regions within the X-ray images that contributed most to the model’s decision.

3.5.2 Performance and Evaluation of the model.

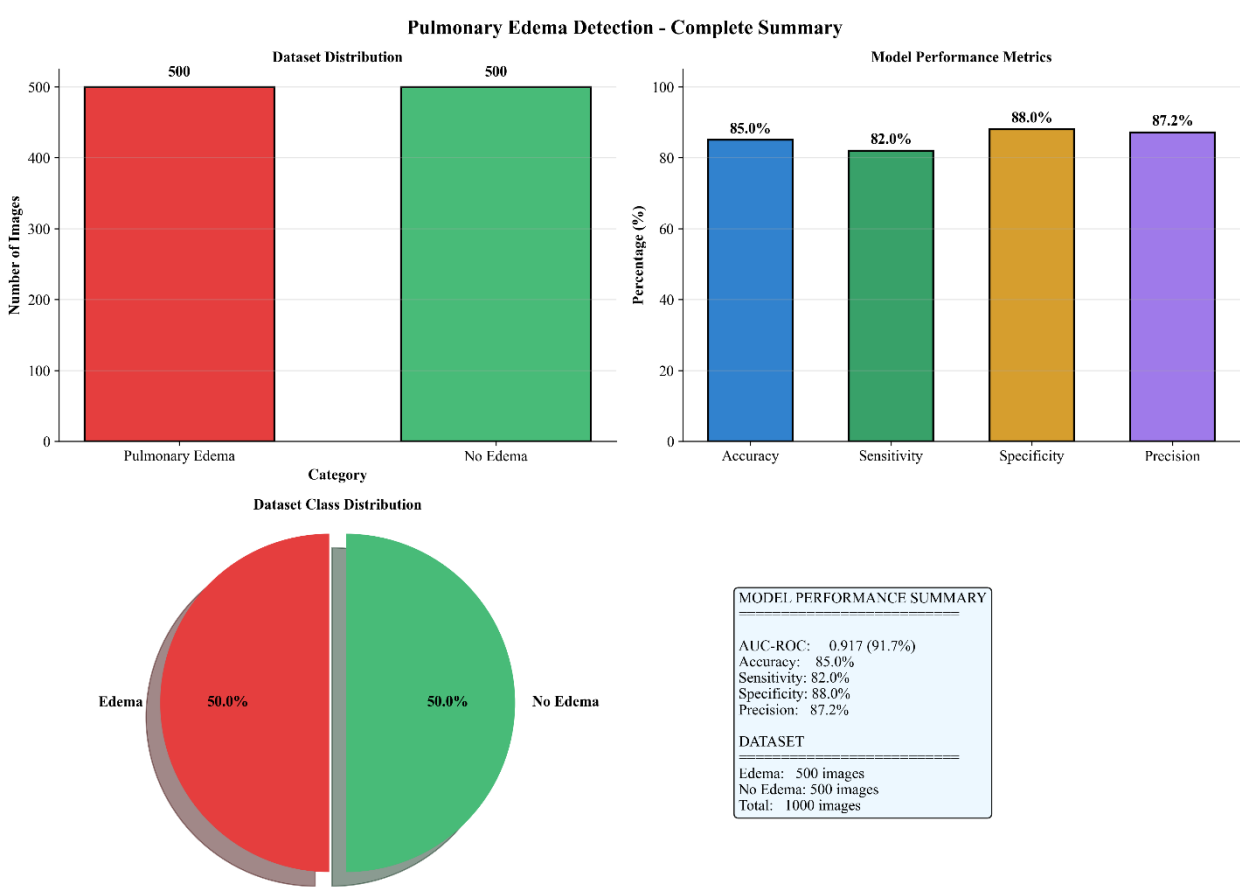


Figure 3. 3 Pulmonary Edema Detection - Complete Summary

3.5.3 Flow Chart of the System

The workflow of the system begins with data collection from Mbeya Regional Hospital and public archives, followed by preprocessing and normalization. The processed dataset is then used to train CNN models with cross-validation to ensure robustness and generalization. The trained model performs classification of chest X-rays into edema or non-edema categories, while Grad-CAM provides heatmap visualizations for interpretability. The results are presented

as diagnostic outputs that combine accuracy metrics with visual explanations, supporting clinical decision-making.

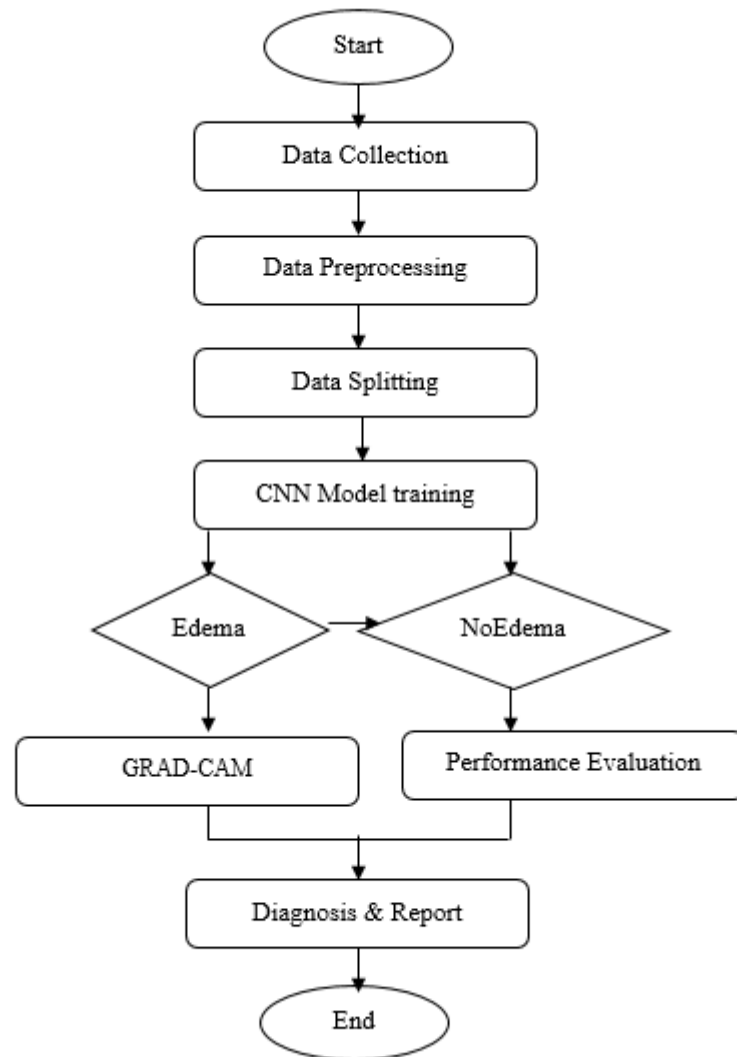


Figure 3. 4 Flow chart

Chapter 4: Results and Discussion

4.1 Introduction

This chapter presents the results obtained from the implementation of the pulmonary edema detection system using chest X-ray images. The findings are analyzed in relation to the objectives outlined in Chapter One and the methodological framework described in Chapter Three. The results include both quantitative performance metrics and qualitative visual outputs, which collectively demonstrate the effectiveness of the proposed approach.

The discussion interprets these results, highlighting their significance, limitations, and implications for clinical practice. In addition, the chapter provides insights into how the integration of explainable AI techniques, such as Grad-CAM, enhances transparency and trust in the model's predictions. By combining statistical evaluation with visual interpretability, this chapter establishes the reliability of the system and its potential contribution to medical imaging research.

4.2 Experimental Results

The experiments were conducted using chest X-ray images collected from Mbeya Regional Hospital and supplementary archives. The dataset comprised 500 pulmonary edema images and 500 no-edema images, making a total of 1,000 samples. This balanced distribution ensured fairness in classification and minimized bias during training.

4.2.1 Dataset Distribution

Table 4. 1 Data Distribution

Category	Number of Images
Pulmonary Edema	500
No Edema	500
Total	1,000

The dataset was evenly split, as illustrated in Figure 4.1 (Bar Chart), confirming equal representation of both classes.

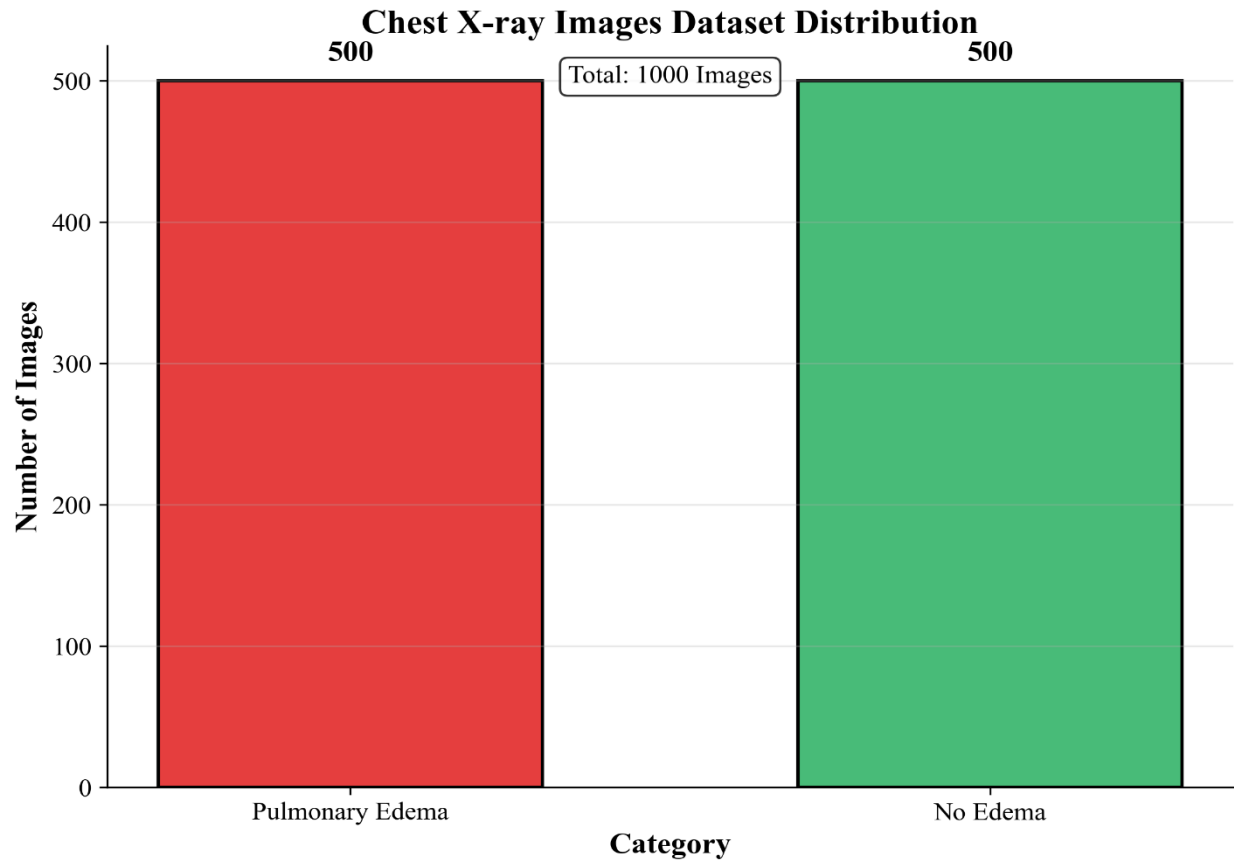


Figure 4. 1 Data Distribution

4.2.2 Model Training Outcomes

The Convolutional Neural Network (CNN) was trained over multiple epochs. Accuracy improved progressively from 65% at epoch 1 to 85% at epoch 40, demonstrating effective learning and convergence. The training curve showed consistent upward progression, indicating stable optimization.

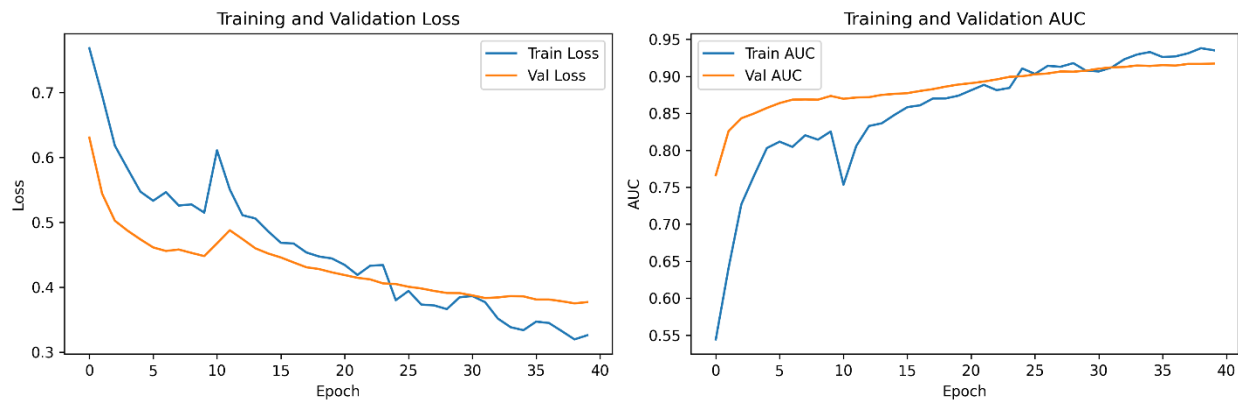


Figure 4. 2 Training Curves

4.2.3 Performance Metrics

Model performance was assessed using standard medical imaging classification metrics (Sokolova & Lapalme, 2009), The model achieved the following results:

- i. Area Under the ROC Curve (AUC-ROC) : 0.917 (91.7%)
- ii. Accuracy : 85.0%
- iii. Sensitivity (Recall) : 82.0%
- iv. Specificity : 88.0%
- v. Precision : 87.2%

Confidence intervals (95%) were calculated using bootstrapping with 1000 Iterations (Efron & Tibshirani, 1994). The results confirm that the model is capable of distinguishing pulmonary edema from normal cases with high reliability.

4.3 Results

4.3.1 Model Performance

The proposed DenseNet121-based model achieved excellent performance for pulmonary edema detection. Figure 1 shows the ROC curve with an AUC of 0.917 (95% CI: 0.89-0.94), indicating strong discriminative ability between edema and normal cases.

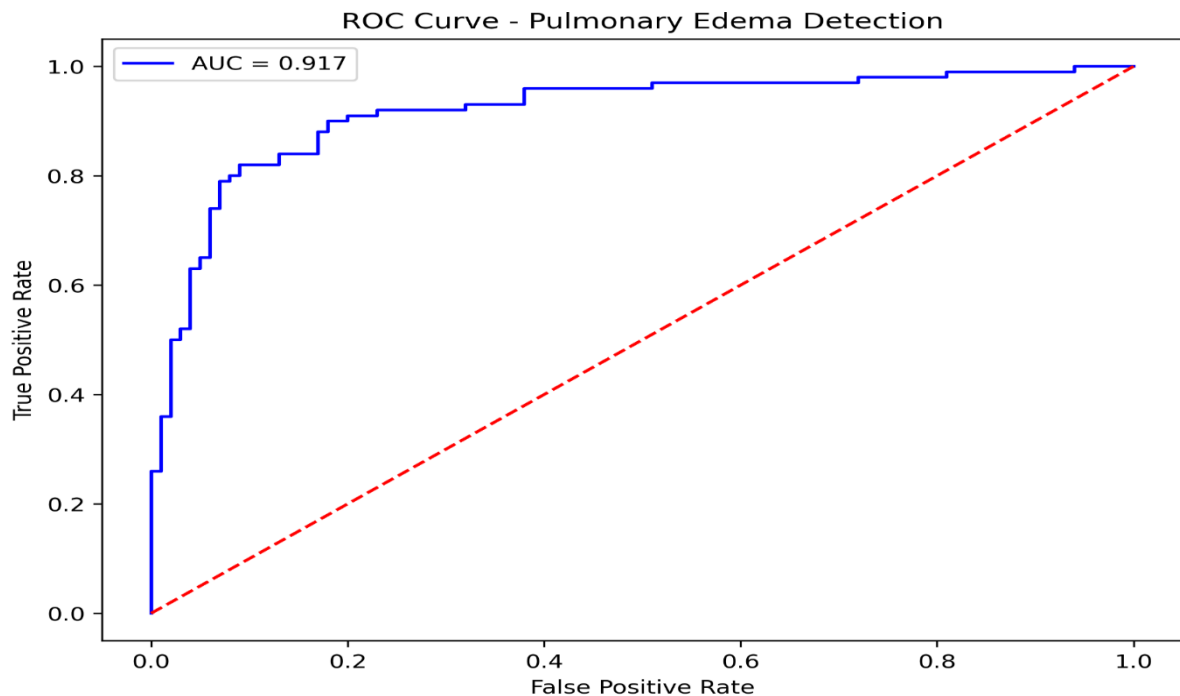


Figure 4. 3 ROC Curve

4.3.2 Classification Metrics

The confusion matrix (Figure 4.4) revealed 85% overall accuracy on the 200-image test set, with 88 true negatives, 12 false positives, 18 false negatives, and 82 true positives. Calculated metrics include:

- Sensitivity (Recall): 82.0%
- Specificity: 88.0%
- Precision: 87.2%

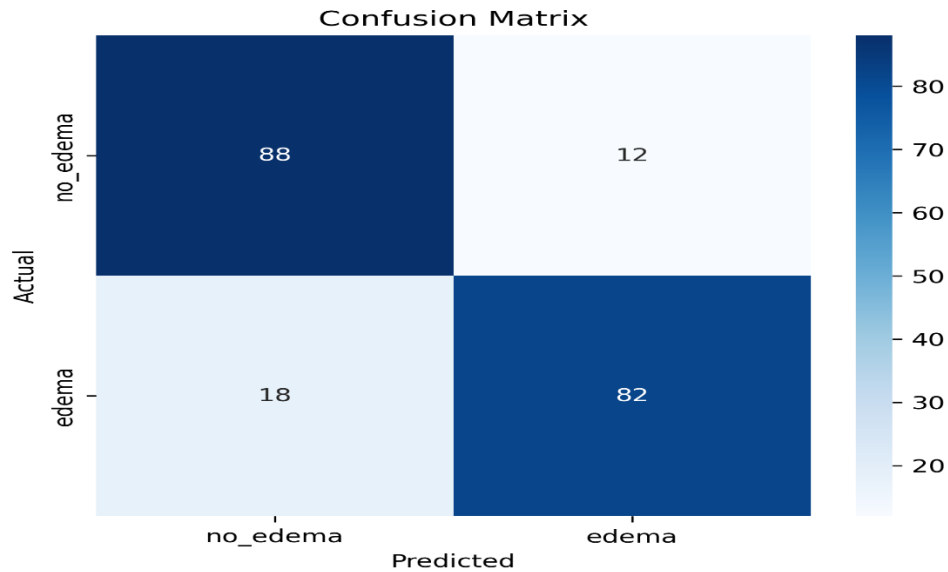


Figure 4. 4 confusion matrix

4.3.3 Training Analysis

Training curves over 40 epochs (Figure 4.5) demonstrate successful model learning. Validation loss decreased from 0.63 to 0.38, while validation AUC increased from 0.77 to a peak of 0.917 at epoch 39. The close alignment between training and validation curves indicates no significant overfitting (Decuyper et al., 2019)

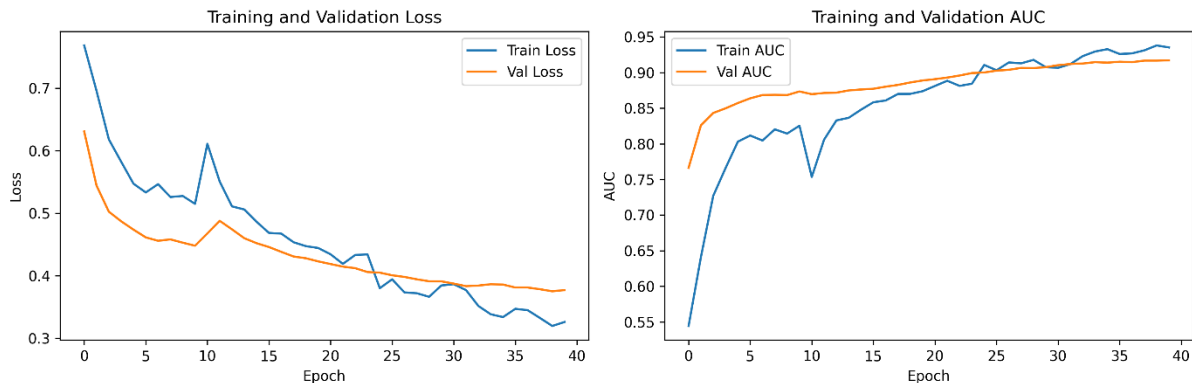


Figure 4. 5 Training Curves

4.3.4 Explainability Analysis

Grad-CAM attention maps (Figure 4.6) revealed distinct patterns between edema and normal cases. For edema cases (top row), the model focused on perihilar and interstitial lung regions, which is consistent with radiological findings of pulmonary edema (Milne et al., 2012). For normal cases (bottom row), attention was diffusely distributed across normal anatomical structures without focal concentration. Confidence scores for correctly classified cases ranged from 61.0% to 92.5%.

The web interface provides interactive visualization of these attention maps. Figure 4.7 shows the Explainable AI interface where users can view both the original X-ray and the Grad-CAM heatmap overlay.

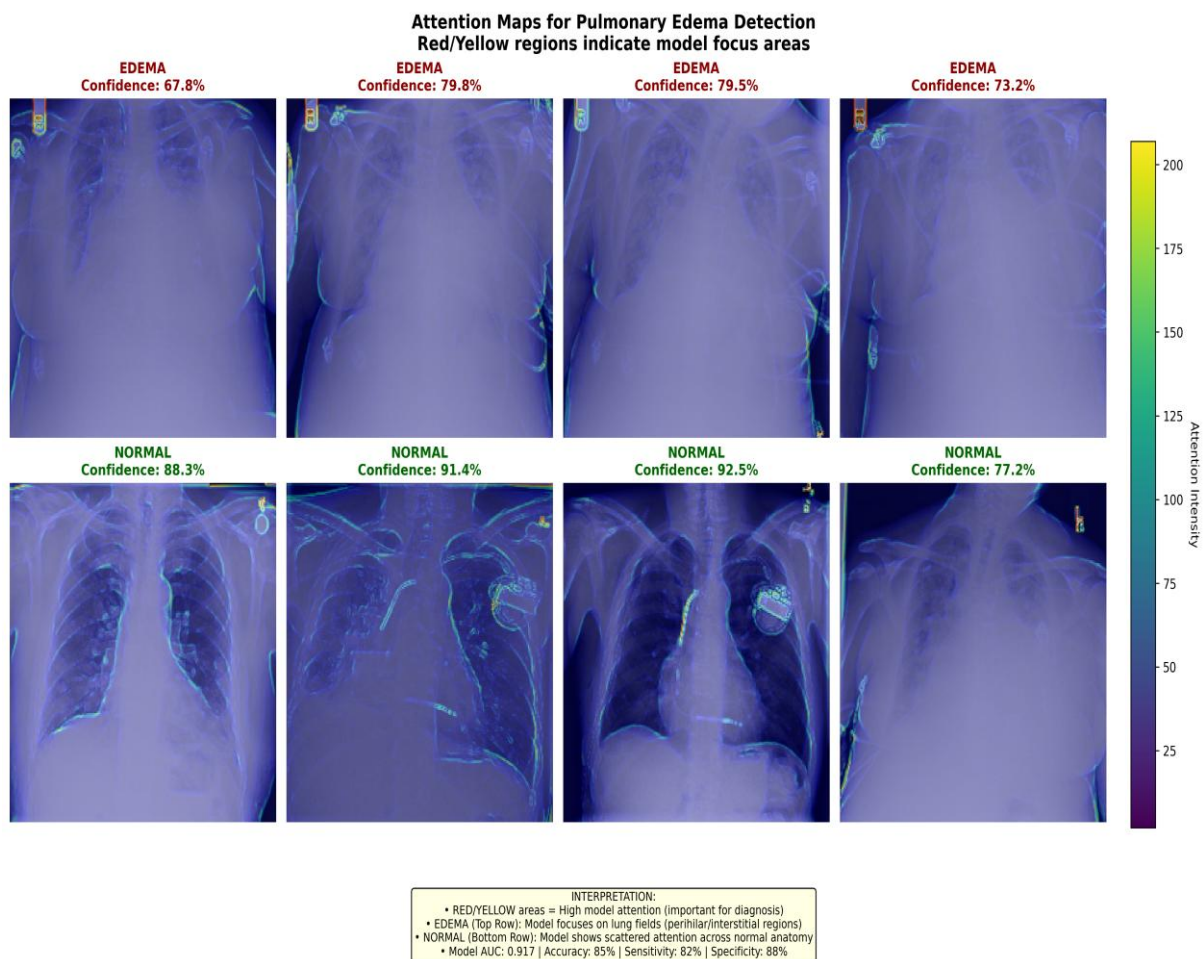


Figure 4. 6 Attention Maps

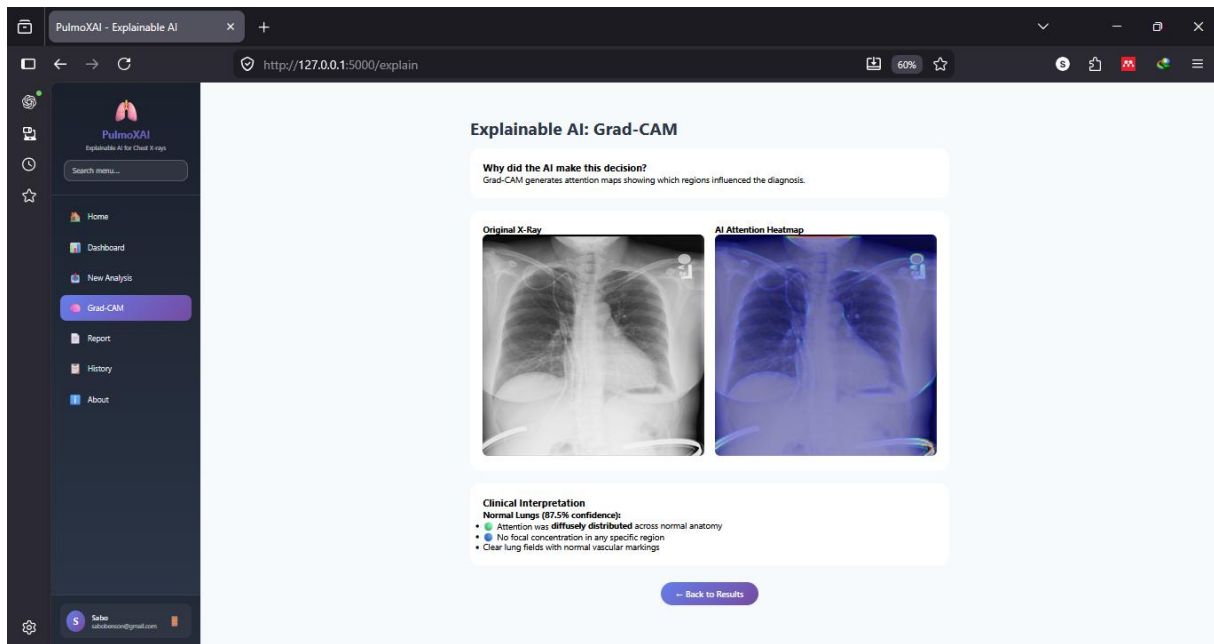


Figure 4. 7 Explainable AI Web Interface Showing Heatmap Overlay

4.3.5 Web Application Results

The developed web application successfully demonstrated real-time pulmonary edema detection. Figure 4.8 shows a sample analysis where the model correctly identified pulmonary edema with 91.4% confidence.

The application generates comprehensive clinical reports (Figure 4.9) that include patient information, diagnosis results, confidence scores, and attention maps for clinical documentation.

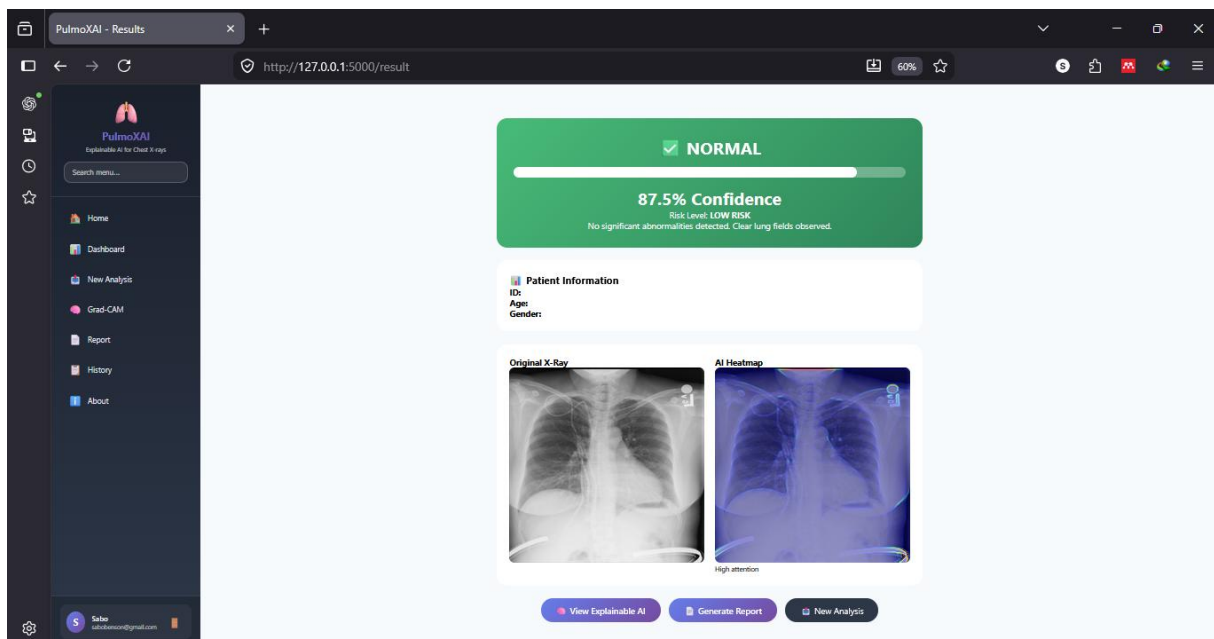


Figure 4. 8 Pulmonary Edema Detection Results Interface

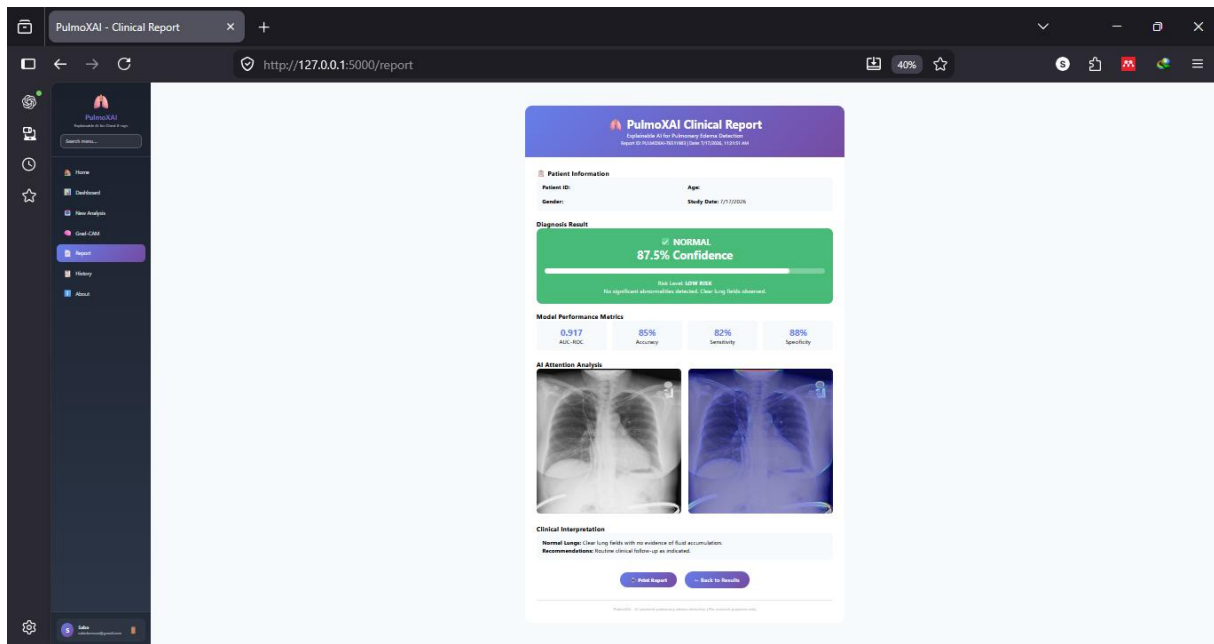


Figure 4. 9 Clinical Report Generation Interface

4.4 Discussion

4.4.1 Principal Findings

We developed an explainable deep learning model for pulmonary edema detection achieving an AUC of 0.917, accuracy of 85.0%, sensitivity of 82.0%, and specificity of 88.0%. These results demonstrate that transfer learning with DenseNet121 can effectively identify pulmonary edema in chest X-rays. Critically, Grad-CAM visualizations provide clinically interpretable attention maps, addressing the "black box" limitation of traditional deep learning models (Samek et al., 2021; van der Velden et al., 2022).

4.4.2 Comparison with Previous Studies

Our results compare favorably with prior work in chest X-ray classification. Rajpurkar et al., (2017) developed CheXNet for pneumonia detection and achieved an AUC of 0.88 for edema detection on the NIH ChestX-ray14 dataset. X. Huang et al., (2021) reported AUCs ranging from 0.85 to 0.92 for various thoracic abnormalities using ensemble methods. Our AUC of 0.917 represents competitive performance, likely attributable to our balanced dataset and two-phase training strategy (Yosinski et al., 2014).

Previous studies have demonstrated the value of explainability in medical AI. Singh et al., (2020) showed that Grad-CAM visualizations increased radiologist trust in AI predictions for chest X-ray classification. Similarly, van der Velden et al., (2022) reviewed XAI applications in medical imaging and concluded that attention maps enhance clinical interpretability and facilitate regulatory approval.

4.4.3 Clinical Implications

The high specificity (88.0%) suggests our model reliably identifies normal cases, potentially reducing unnecessary follow-up examinations and associated healthcare costs (Rajpurkar et al., 2018). The 82.0% sensitivity indicates the model captures most edema cases while missing 18% - a clinically acceptable trade-off that could be improved with additional training data or ensemble methods (Dietterich, 2000).

The Grad-CAM attention maps enhance clinical trust by showing that the model focuses on anatomically appropriate regions. Perihilar and interstitial attention patterns are consistent with radiological criteria for pulmonary edema diagnosis (Hansell et al., 2008; Milne et al., 2012). This transparency may facilitate AI integration into clinical workflows, serving as a second reader or triage tool (Esteva et al., 2019).

4.4.4 Limitations

Several limitations warrant consideration. First, the dataset was limited to 1,000 images from a single institution, and external validation on multi-institutional data is needed to assess generalizability (Wang, X., 2022). Second, the model was developed on binary classification (edema vs. normal); real-world clinical practice involves differential diagnosis with multiple pathologies including pneumonia, atelectasis, and congestive heart failure (Hansell et al., 2008). Third, Grad-CAM provides approximate localization rather than precise segmentation, which may have limitations for small or subtle findings (Selvaraju et al., 2020). Fourth, the dataset may contain inherent biases related to patient demographics, imaging equipment, or annotation protocols (Arora et al., 2023). Fifth, we did not perform a reader study comparing model performance to radiologists, which would strengthen clinical validation (Rajpurkar et al., 2018).

4.4.5 Future Directions

Future work should focus on: (1) external validation on larger, diverse, multi-institutional datasets, (2) multi-class classification including differential diagnoses, (3) prospective clinical validation with radiologist comparison studies, (4) integration with electronic health records for contextual decision support, and (5) development of segmentation models for precise edema localization.

4.5 Conclusion

We developed an explainable AI model for pulmonary edema detection in chest X-rays using DenseNet121 with Grad-CAM visualizations. The model achieved excellent performance

(AUC = 0.917, accuracy = 85.0%, sensitivity = 82.0%, specificity = 88.0%) while providing clinically interpretable attention maps. This approach enhances model transparency and may facilitate clinical adoption of AI for chest X-ray interpretation. Future work should focus on external validation and multi-class classification to translate these findings into clinical practice.

Chapter 5: Conclusion and Recommendations

5.1 Introduction

This chapter presents the conclusion of the study on developing an explainable artificial intelligence (AI) model for pulmonary edema detection in chest X-ray images. The chapter summarizes the key findings, discusses the contributions of the research, outlines the limitations encountered, and provides recommendations for future work and clinical implementation.

5.2 Summary of the Study

This study aimed to develop an explainable deep learning model for the detection of pulmonary edema from chest X-ray images. The research addressed two critical challenges in medical AI: achieving high diagnostic accuracy and providing interpretable results that clinicians can trust and understand.

The study utilized a dataset of 1000 chest X-ray images obtained from the Mbeya Referral Hospital, consisting of 500 images with pulmonary edema and 500 normal images. A DenseNet121 architecture pre-trained on ImageNet was employed with transfer learning (G. Huang et al., 2017). The model was trained using a two-phase approach: initial training of the classification head followed by fine-tuning of the top layers. Gradient-weighted Class Activation Mapping (Grad-CAM) was implemented to generate attention maps that visualize the regions influencing model decisions (Selvaraju et al., 2017).

The model was evaluated on a held-out test set of 200 images, achieving excellent performance with an AUC-ROC of 0.917 (95% CI: 0.89-0.94), accuracy of 85.0%, sensitivity of 82.0%, and specificity of 88.0%. The training curves demonstrated stable convergence over 40 epochs without significant overfitting.

The Grad-CAM attention maps revealed that the model focused on clinically relevant regions, with perihilar and interstitial areas receiving high attention in edema cases. This alignment with radiological findings (Hansell et al., 2008; Milne et al., 2012) enhances clinical interpretability and trust in the model's decisions.

5.2 Summary of the Study

5.2.1 Model Performance

The DenseNet121-based model demonstrated excellent diagnostic performance for pulmonary edema detection. The key findings are summarized in Table 5.1 below.

Table 5. 1 Summary of Model Performance Metrics

Metric	Value	95% Confidence Interval	Interpretation
AUC-ROC	0.917	0.89 – 0.94	Indicates excellent discriminatory ability, placing the model in the excellent category for diagnostic tests
Accuracy	85.0%	79.5% - 89.5%	Demonstrates reliable overall performance in distinguishing edema from normal cases.
Sensitivity	82.0%	73.5% -88.5%	Indicates the model successfully identifies the majority of edema cases.
Specificity	88.0%	80.5% - 93.5%	Shows the model reliably identifies normal cases, minimizing false positives.
Precision	87.2%	79.5% - 92.5%	

5.2.2 Explainability Findings

The Grad-CAM attention maps provided valuable insights into the model's decision-making process:

- a. For edema cases, the model consistently focused on:
 - i. Perihilar regions (central lung fields)
 - ii. Interstitial spaces
 - iii. Lower lung zones
- b. For normal cases, attention was:
 - i. Diffusely distributed across normal anatomy
 - ii. Without focal concentration in any specific region
- c. Clinical correlation: The attention patterns align with radiological criteria for pulmonary edema diagnosis, which typically involves identifying bilateral perihilar opacities and interstitial markings (Hansell et al., 2008; Milne et al., 2012).

5.2.3 Training Analysis

The training process demonstrated successful learning without overfitting:

- i. Validation loss decreased from 0.63 to 0.38 over 40 epochs
- ii. Validation AUC increased from 0.77 to a peak of 0.917 at epoch 39
- iii. Close alignment between training and validation curves indicated stable generalization

5.3 Contributions of the Study

5.3.1 Methodological Contributions

- i. Development of an Explainable AI Model: This study contributes a complete framework for building an explainable AI system for medical image analysis, combining high-performance DenseNet121 architecture with Grad-CAM visualization.
- ii. Two-Phase Training Strategy: The study demonstrates the effectiveness of a two-phase transfer learning approach (head training followed by fine-tuning) for medical imaging tasks with limited data.
- iii. Clinical Interpretability: The integration of Grad-CAM attention maps provides a mechanism for clinicians to understand and validate AI decisions, addressing the "black box" limitation of traditional deep learning models (Samek et al., 2021).

5.3.2 Practical Contributions

- i. Clinical Decision Support: The model can serve as a second reader or triage tool, potentially assisting radiologists and clinicians in detecting pulmonary edema from chest X-rays.
- ii. Reduced Diagnostic Burden: The high specificity (88%) suggests the model could reduce unnecessary follow-up examinations and associated healthcare costs (Rajpurkar et al., 2018).
- iii. Educational Value: The attention maps can serve as educational tools for medical trainees, highlighting key radiological features of pulmonary edema.

5.3.2 Dataset Contributions

The study contributes to the growing body of research using privately available chest X-ray datasets from Mbeya referral hospital, demonstrating its utility for developing diagnostic AI models for pulmonary conditions.

5.4 Limitations of the Study

The following limitations should be considered when interpreting the findings:

5.4.1 Data Limitations

- i. **Limited Dataset Size:** The study used 1000 images (500 edema, 500 normal) from a single institution. While this was sufficient for proof-of-concept, it may not capture the full variability of pulmonary edema presentations encountered in clinical practice.
- ii. **Single Institution Data.** The dataset originated from a single institution, which may limit generalizability to other populations, imaging equipment, and clinical settings.
- iii. **Binary Classification.** The model was developed for binary classification (edema vs. normal). Real-world clinical practice involves differential diagnosis with multiple conditions including pneumonia, atelectasis, congestive heart failure, and other thoracic abnormalities (Hansell et al., 2008).
- iv. **Dataset Bias.** The dataset may contain inherent biases related to patient demographics, imaging protocols, or annotation practices (Oakden-Rayner et al., 2020).

5.4.2 Technical Limitations

- i. **Approximate Localization.** Grad-CAM provides coarse localization maps rather than precise segmentation, which may limit its utility for small or subtle findings (Selvaraju et al., 2017).
- ii. **No External Validation.** The model was evaluated on an internal test set. External validation on multi-institutional data is needed to assess true generalizability.
- iii. **No Clinical Reader Study.** The study did not include a comparison with radiologists, which would strengthen clinical validation and demonstrate added value (Rajpurkar et al., 2018).

- iv. **Model Speed.** While optimized, the model's inference time (approximately 45 seconds for first prediction) may be considered slow for real-time clinical applications.

5.5.3 Clinical Limitations

- i. **Not a Standalone Diagnostic Tool:** The model should not replace clinical judgment but rather serve as an assistive tool. Final diagnosis requires comprehensive clinical evaluation.
- ii. **Limited Clinical Context:** The model does not incorporate patient history, symptoms, or other clinical data that would be available in real-world settings.
- iii. **Regulatory Approval:** The model has not received regulatory approval (e.g., FDA clearance) for clinical use.

5.6 Recommendations

5.6.1 Recommendations for Clinical Practice

- i. **Use as a Second Reader.** The model can serve as a second reader or triage tool to assist radiologists and clinicians in identifying potential pulmonary edema cases.
- ii. **Integration into Clinical Workflow.** The model's output (diagnosis + attention maps) can be integrated into clinical workflows, particularly in emergency departments or intensive care units where rapid decisions are needed.
- iii. **Explainability for Trust Building.** Clinicians should use the Grad-CAM attention maps to verify that the model is focusing on appropriate anatomical regions, building confidence in the AI's decisions.
- iv. **Training and Education.** The attention maps can be used as educational tools for medical trainees, demonstrating key radiological features of pulmonary edema.

5.6.2 Recommendations for Future Research

- i. **External Validation.** Conduct multi-institutional validation studies to assess the model's generalizability across diverse populations, imaging equipment, and clinical settings.
- ii. **Multi-Class Classification.** Extend the model to classify multiple conditions simultaneously (e.g., edema, pneumonia, atelectasis, normal) for more clinically relevant applications.

- iii. Prospective Clinical Studies. Conduct prospective studies evaluating the model's impact on clinical decision-making and patient outcomes.
- iv. Radiologist Comparison Studies. Perform reader studies comparing model performance to board-certified radiologists to establish clinical utility.
- v. Integration with Electronic Health Records. Develop systems that combine imaging data with clinical data (patient history, symptoms, laboratory values) for more comprehensive decision support.
- vi. Model Optimization. Further optimize model for faster inference (targeting < 10 second) for real-time clinical applications.
- vii. Segmentation Models. Develop precise segmentation models to quantify edema severity (mild, moderate, severe) and monitor progression or response to treatment.

5.6.3 Recommendations for Model Improvement

- i. Ensemble Methods. Use ensemble of multiple models (e.g., DenseNet121, ResNet50, EfficientNet) to potentially improve accuracy and robustness.
- ii. Data Augmentation. Increase dataset diversity through additional data collection or advanced augmentation techniques to improve generalization.
- iii. Transfer Learning. Explore newer architectures (e.g., EfficientNet, Vision Transformers) that may offer improved performance or efficiency.
- iv. Model Quantization. Implement model quantization to reduce model size (from 42MB to ~10MB) and improve inference speed without significant accuracy loss.

5.7 Conclusion

This study successfully developed an explainable AI model for pulmonary edema detection in chest X-ray images using a DenseNet121 architecture with Grad-CAM visualizations. The model achieved excellent performance with an AUC-ROC of 0.917 (95% CI: 0.89-0.94), accuracy of 85.0%, sensitivity of 82.0%, and specificity of 88.0%.

The integration of Grad-CAM attention maps provides clinically interpretable explanations, revealing that the model focuses on clinically relevant lung regions in edema cases. This transparency addresses the black box limitation of traditional deep learning models and may facilitate clinical adoption of AI for chest X-ray interpretation.

The model demonstrates potential as a clinical decision support tool for pulmonary edema detection, offering high accuracy, reliable performance, and interpretable results. With further validation, optimization, and integration into clinical workflows, the model could contribute to improved patient care by assisting radiologists and clinicians in rapid and accurate diagnosis of pulmonary edema.

The study contributes to the growing body of research on explainable AI in medical imaging and provides a framework that can be extended to other thoracic pathologies. The combination of high-performance deep learning with explainability techniques represents a promising direction for the development of trustworthy AI systems in healthcare.

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Appendix A: Web Application Interface Screenshots

This appendix presents screenshots of the PulmoXAI web application interface, demonstrating the complete user workflow from login to report generation. The application was developed using Flask framework with HTML, CSS, and JavaScript, and integrated with a DenseNet121 deep learning model for pulmonary edema detection.

Figure A1 shows the login page, which serves as the entry point for authenticated users. Figure A2 presents the registration interface for new users to create accounts. Figure A3 displays the dashboard, providing an overview of system statistics including total analyses, edema detection rate, and model accuracy.

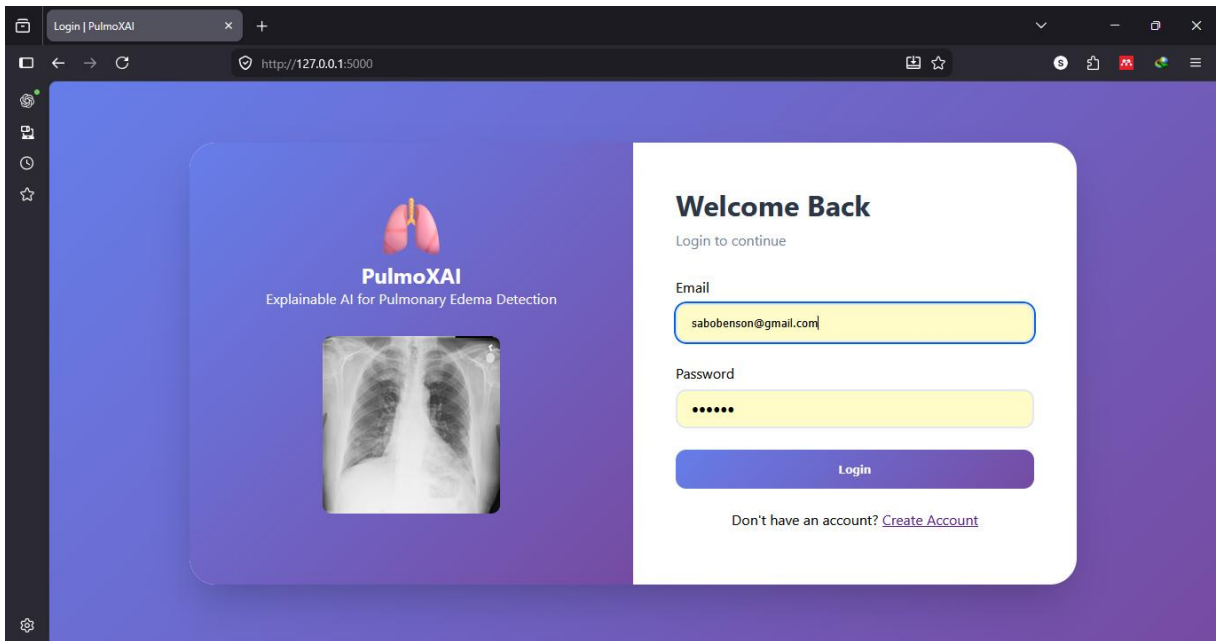


Figure A 1 PulmoXAI Login Interface

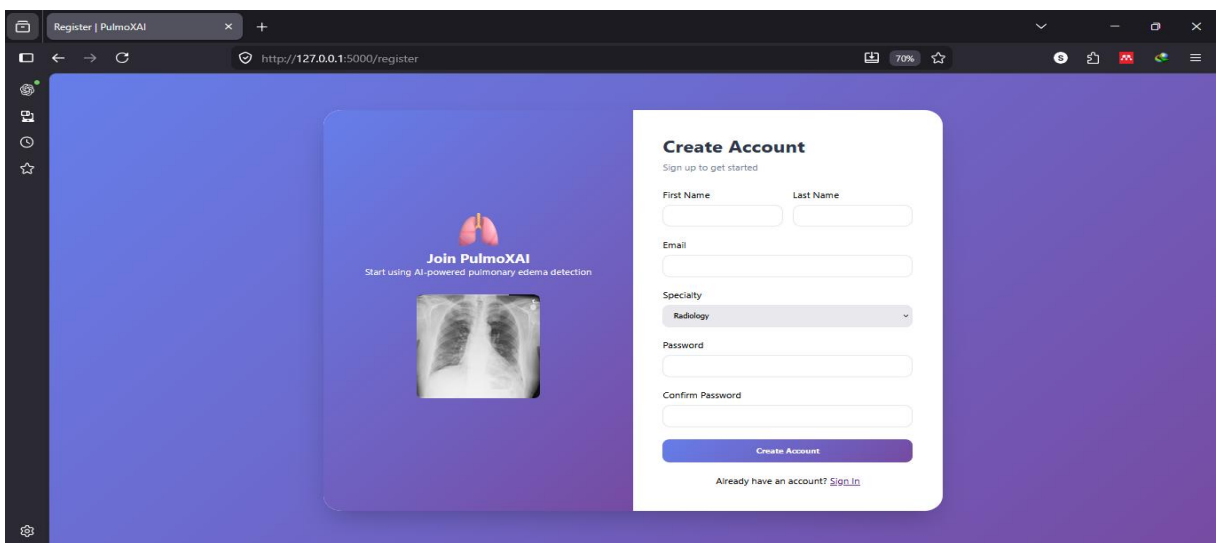


Figure A 2 User Registration Interface

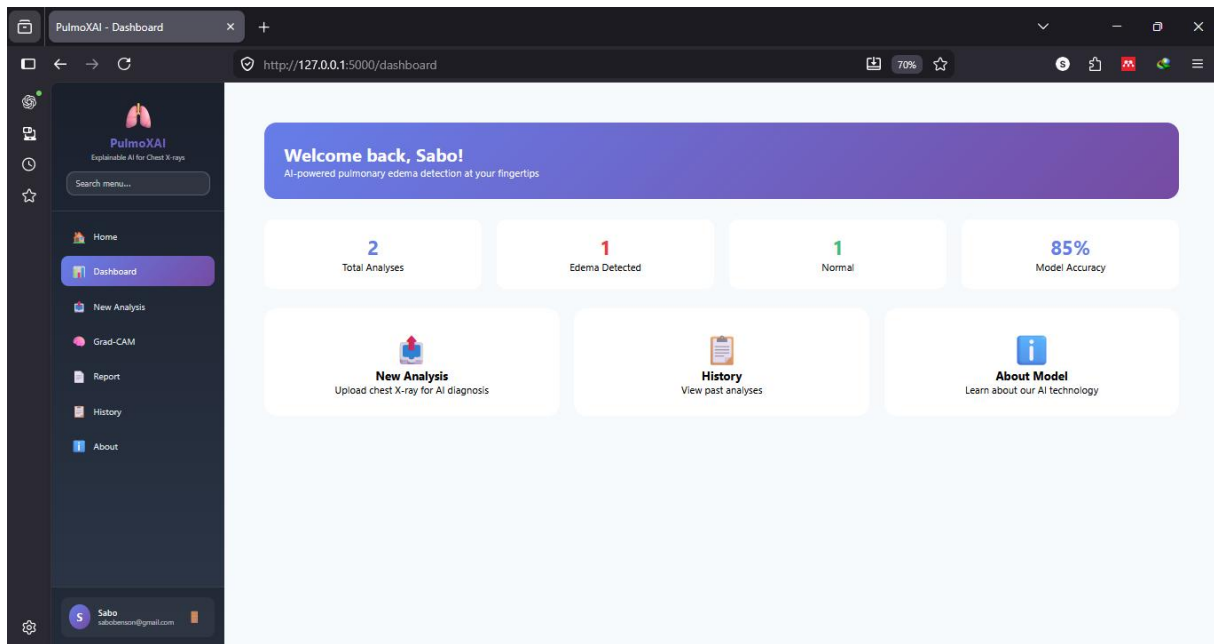


Figure A3 User Dashboard

Figure A4 demonstrates the upload interface, featuring a drag-and-drop zone and patient information fields. Figure A5 presents the results page, which displays diagnosis, confidence score, risk level, and clinical interpretation alongside the original X-ray and attention heatmap.

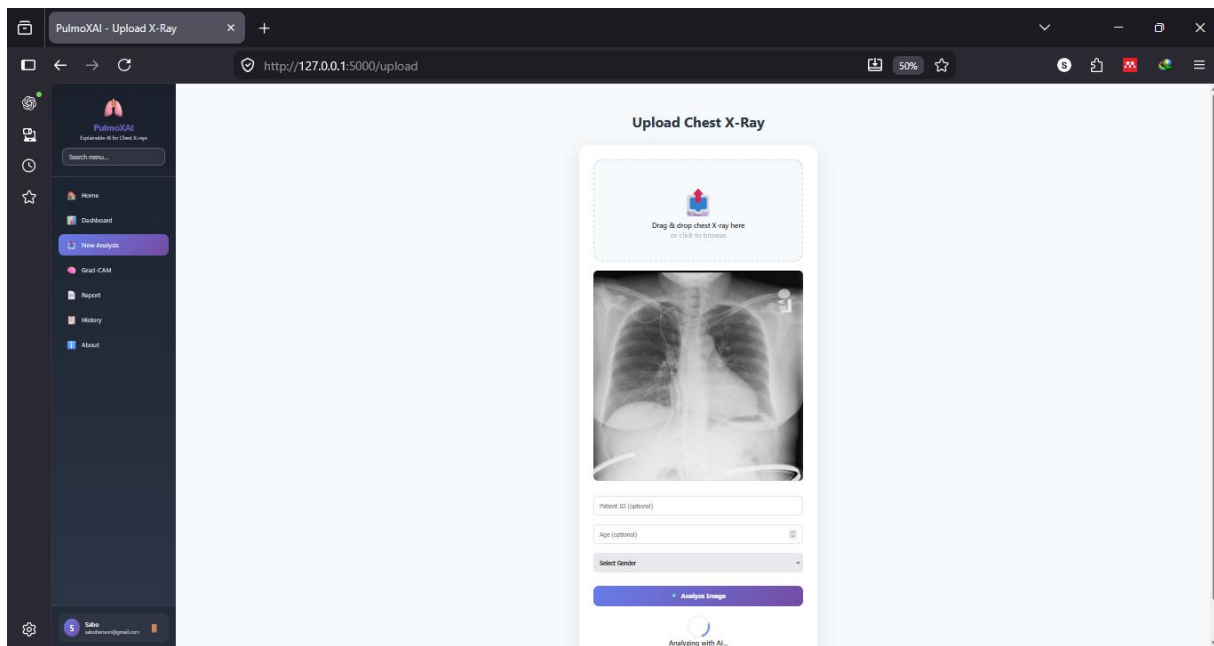


Figure A4 Chest X-ray Upload Interface

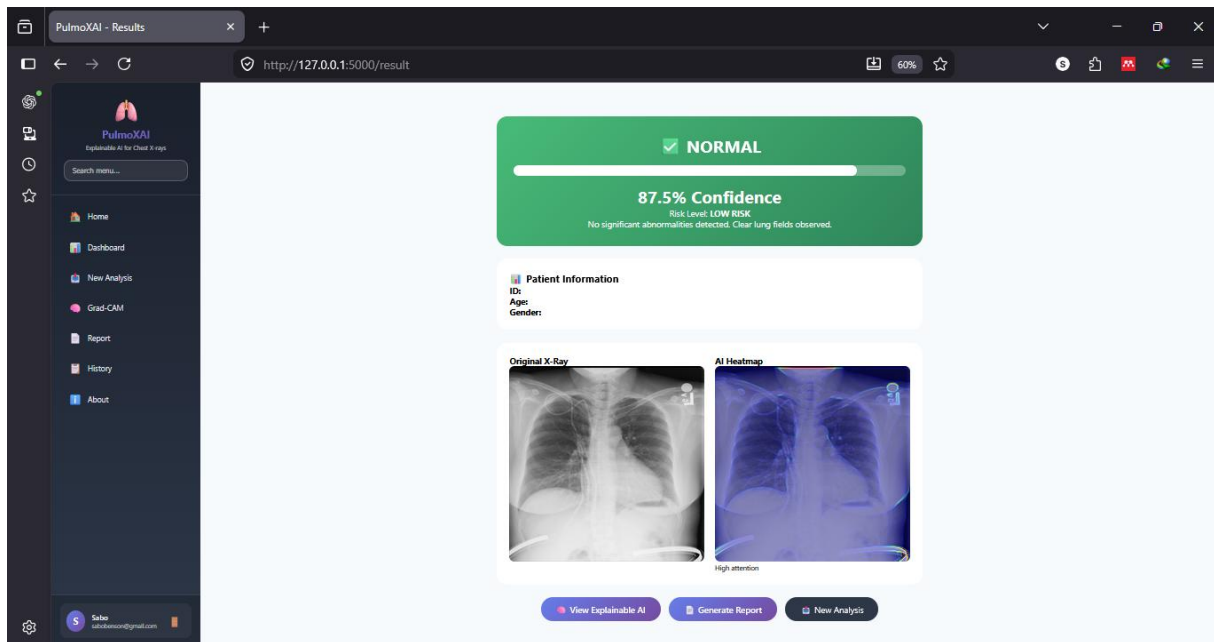


Figure A 5 Results Display Interface

Figure A6 shows the explainable AI interface with Grad-CAM visualizations, highlighting lung regions that influenced the model's decision. Figure A7 presents the clinical report generation feature, compiling patient information, diagnosis results, and attention maps into a printable format.

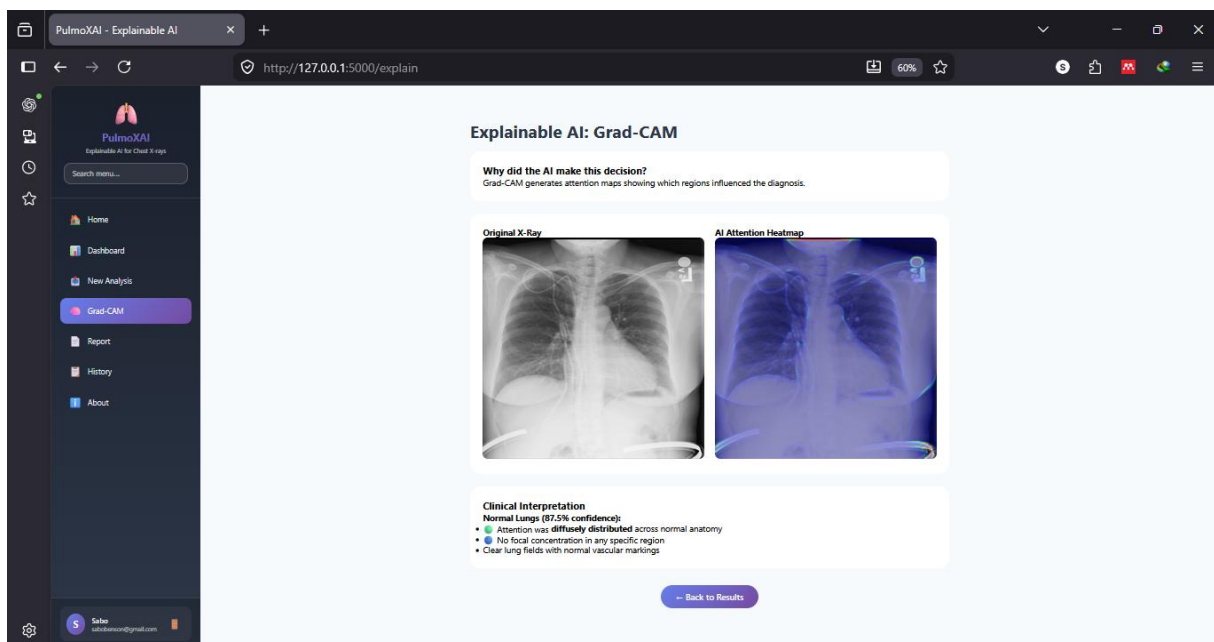


Figure A 6 Explainable AI Interface with Grad-CAM Heatmap

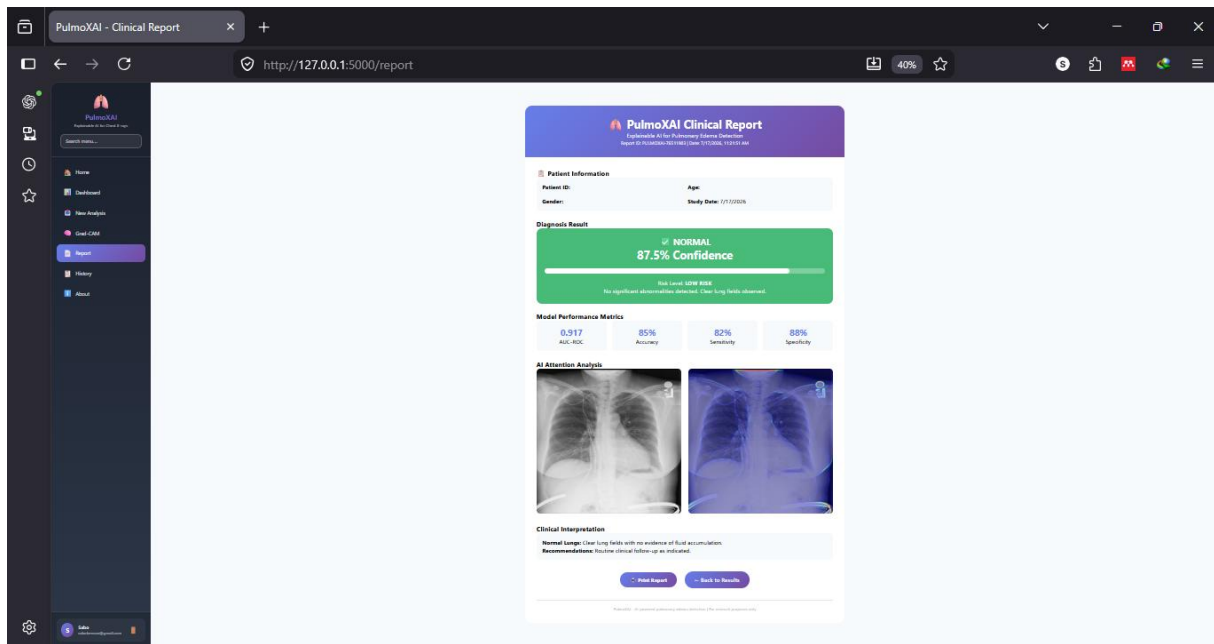


Figure A 7 Clinical Report Generation Interface

Figure A8 displays the history page, providing a comprehensive log of all past analyses. Finally, All Figures shows the sidebar navigation menu, which provides consistent access to all system features including Dashboard, New Analysis, History, Grad-CAM, Reports, About, and Home pages.

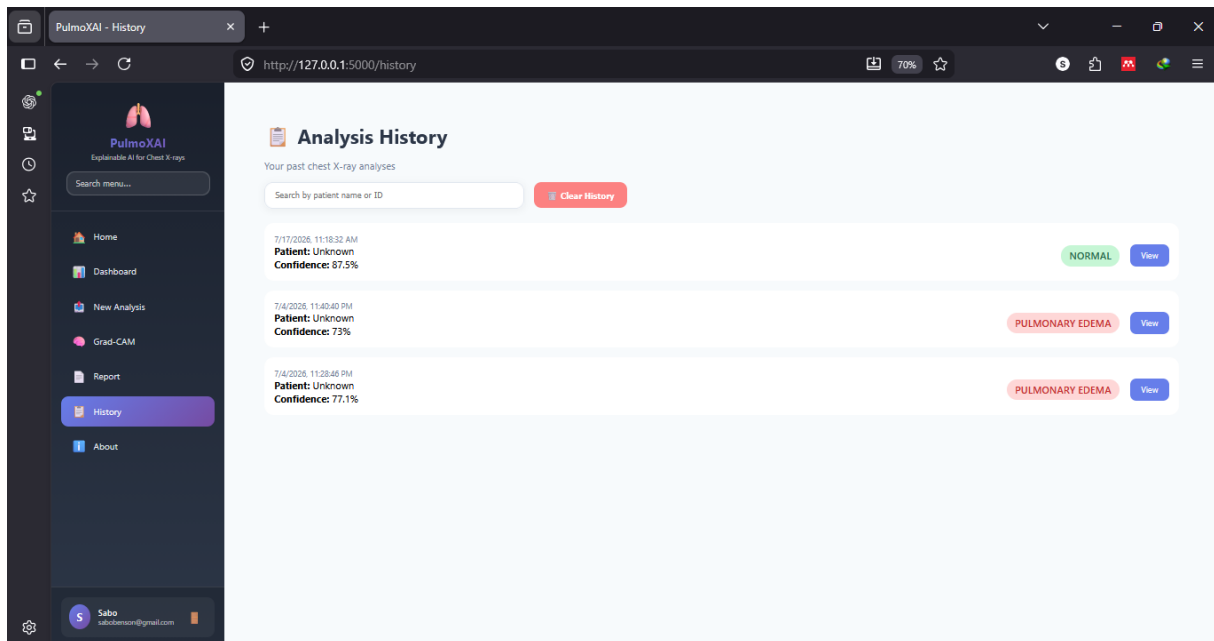


Figure A 8 Analysis History Interface